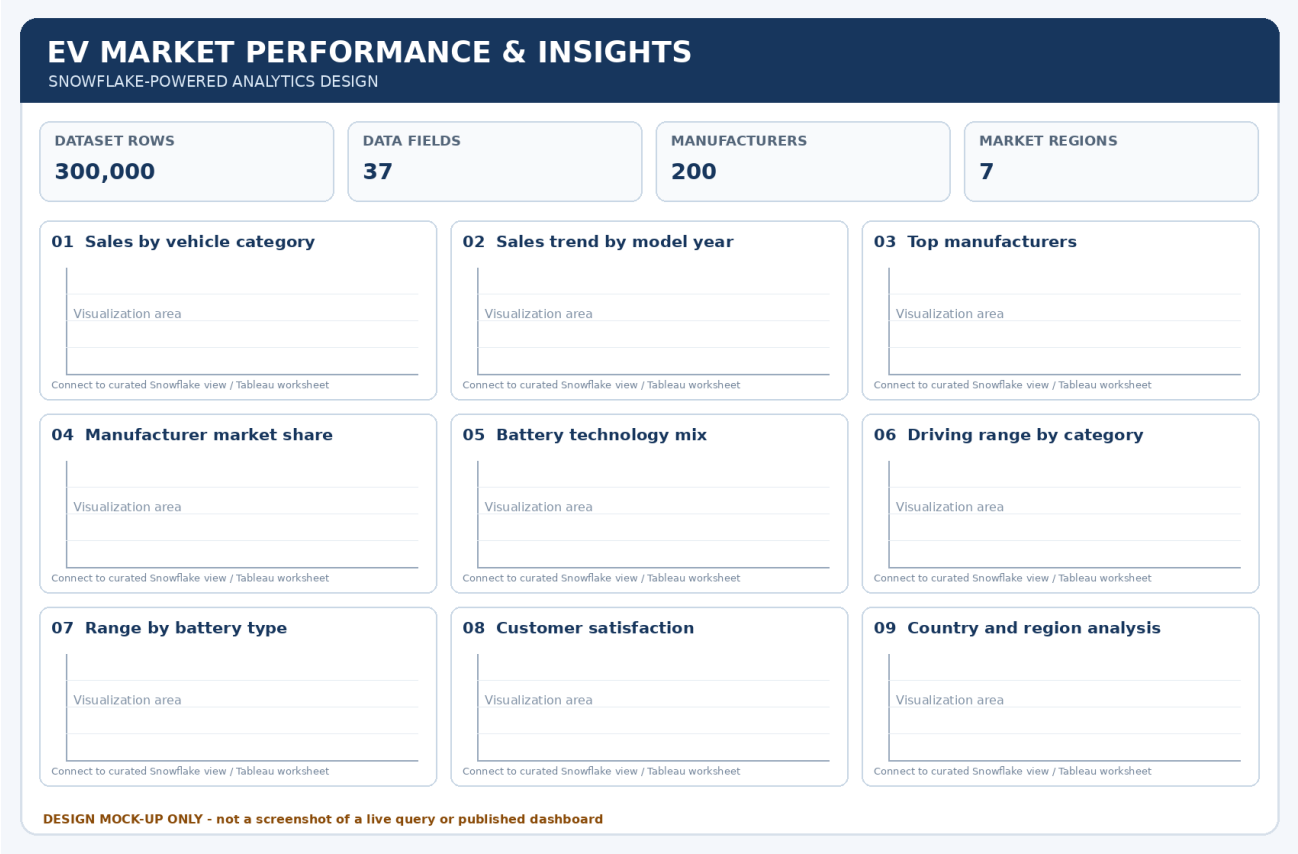
Dashboard Design - Snowflake / Streamlit

Date	29 October 2026
Team ID	Individual Project - Ujjawal Rathod
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	5 Marks

Dashboard goal and audience

Provide a compact, interactive view of EV category demand, model-year trends, competitive fragmentation, battery adoption, driving range, customer satisfaction, and geography. This is a Snowflake-backed design concept for the data/analytics layer; the previously published project dashboard is in Tableau Public.

The following is a design wireframe, not a screenshot of a running Streamlit application or a live Snowflake result. KPI cards use dataset facts from the earlier project handoff; analytical chart panels should be connected to validated MART views.



Layout and interaction specification

Area	Design choice	Purpose
KPI row	Dataset rows, columns, manufacturer count, market-region count	Establish the scope of the analysis without implying a live query was run.
Sales block	Category sales bar; model-year line; regional sales comparison	Identify leading categories and demand trends.
Competition block	Top manufacturers; market-share percentage	Show fragmentation using a stable all-market denominator.
Technology block	Battery share; range by category; range vs. energy consumption	Compare technology mix and performance trade-offs.
Customer / geography block	Rating by category; manufacturing country x market region heatmap	Support satisfaction and supply/geography analysis.
Filters	Model year, market region, vehicle category, battery type, manufacturer	Allow focused exploration; keep filter behavior consistent across charts.
Accessibility	Readable labels, color-blind-safe palette, tooltips, responsive single-column layout on narrow screens	Make the dashboard interpretable and robust.

Major outcomes to validate after implementation

- The prior project summary identified Two-Wheeler Scooter as a leading sales category; confirm the rank using the curated dataset.
- The manufacturer view should demonstrate fragmentation instead of exaggerating the leader by normalizing only the visible top ten.
- Battery technology mix should show NMC and LFP as the two largest groups in the prior summary.
- Driving range is expected to vary strongly by vehicle category; compare category and chemistry together.
- Customer satisfaction should be evaluated using the same 1-5 rating scale across categories.
- The dashboard should expose region and model-year filters while preserving consistent KPI definitions.

Data connections

Suggested Streamlit connection: use the Snowflake Python connector or Snowpark session and query EV_MARKET_DB.MART views. Do not embed passwords in the app; use Streamlit secrets or the deployment platform’s environment variables.